Shashi Kumar

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Summary

Fourth-year Ph.D. student at EPFL & Idiap Research Institute working on post-training and efficient adaptation of LLMs and SpeechLLMs, as well as end-to-end and streaming speech recognition models. First-author publications at NeurIPS, EMNLP, ICASSP, and Interspeech, and a Best Paper Award. Before the Ph.D., I worked in industry training and deploying production ASR systems at scale (120K+ audio hours/month).

Research Focus

My research aims to improve the reasoning, efficiency, and robustness of LLMs and multimodal LLMs, including SpeechLLMs. Key directions include latent reasoning, RL-based post-training with GRPO, test-time compute scaling with looped Transformers, parameter-efficient fine-tuning with FVAE-LoRA, and optimal-transport-based cross-modal alignment with OTTC. Other work covers multitask learning, cross-modal transfer, and streaming ASR. Most recently, I am working on interpretability of cross-modal interactions.

Education

Aug 2023–Present **Doctor of Philosophy (Ph.D.)**
École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland &
Idiap Research Institute, Martigny, Switzerland
Advisors: Prof. Andrea Cavallaro, Prof. Petr Motlíček; expected graduation: Aug 2027.
Industry collaboration with Uniphore, USA, on SpeechLLMs.

2013–2017 **Bachelor of Technology (B. Tech.)**
Indian Institute of Technology (IIT) Guwahati, India
Subject: Electronics and Communication Engineering (ECE)

Selected Publications

For a full publication list, see my [Google Scholar] profile. *equal contribution.

2026 **S Kumar***, Y Kaloga*, P Motlicek, I Kodrasi, and A Cavallaro. "Geometric Latent Reasoning Induces Shorter Generations in LLMs," NeurIPS, 2026. [Paper] [website]
Framed latent reasoning as a geometric path in the LLM's token-embedding space, with chain-of-thought traces as anchors. On math benchmarks with Qwen3, GLR needs about 80% fewer generation steps and runs 5.3× faster than explicit chain-of-thought.

2026 **S Kumar***, Y Labrak*, et al. "When Synthetic Speech Is All You Have: Better Call GRPO," IEEE SLT, 2026. [Paper] [website]
Applied reinforcement learning with GRPO to SpeechLLMs adapted on synthetic speech only. WER drops by 40% relative to supervised fine-tuning, and by 45% when GRPO follows SFT.

2026 Y Kaloga*, **S Kumar***, S A Sheikh*, D Khalil, P Motlicek, and I Kodrasi. "Test-Time Compute Scaling for ASR with Depth-Conditioned Looped Transformers," arXiv preprint arXiv:2606.04678. [Paper] [code]
Introduced LARM, a looped Transformer encoder that reuses one shared block across recurrent passes, so encoder depth can be scaled at inference time. On LibriSpeech, it outperforms a 16-block encoder with nearly 4× fewer parameters.

2025 **S Kumar**, Y Kaloga, J Mitros, P Motlicek, and I Kodrasi. "Latent Space Factorization in LoRA," NeurIPS, 2025. [Paper] [code]

Proposed FVAE-LoRA, a parameter-efficient fine-tuning method that separates task-relevant and residual features, improving robustness and performance across image, text, and audio tasks.

2025 Y Kaloga*, **S Kumar***, P Motlicek, and I Kodrasi. "A Differentiable Alignment Framework for Sequence-to-Sequence Modeling via Optimal Transport," arXiv preprint arXiv:2502.01588. [Paper] [code]

Introduced a differentiable sequence-alignment framework based on 1D optimal transport, enabling end-to-end ASR with a single learned alignment.

2025 **S Kumar**, et al. "Performance Evaluation of SLAM-ASR: The Good, the Bad, the Ugly, and the Way Forward," SALMA Workshop, ICASSP, 2025. [Paper] **[Best Paper Award]**

Evaluated SLAM-ASR across domain shifts and speech perturbations, identifying major robustness gaps and providing practical guidance for building more reliable LLM-based ASR systems.

2025 **S Kumar**, et al. "XLSR-Transducer: Streaming ASR for Self-Supervised Pretrained Models," ICASSP, 2025. [Paper]

Enabled streaming ASR with self-supervised encoders, outperforming a streaming Zipformer by 42% relative WER. Attention sinks halve the left context while improving WER by 12% relative.

2024 **S Kumar**, et al. "TokenVerse: Towards Unifying Speech and NLP Tasks via Transducer-based ASR," EMNLP, 2024. [Paper]

Proposed a transducer-based framework that unifies transcription, speaker change detection, end-pointing, and named entity recognition in a single model, outperforming cascaded pipelines.

Experience

May 2022–Apr 2023 **Senior ML Engineer** → **Lead ML Engineer**, *Level AI*, Remote, California, USA

- Led low-resource conversational ASR using domain adaptation and entity-aware transducers.
- Built the production VAD and speaker diarization pipeline, outperforming the vendor system.
- Scaled ASR deployment to 120K+ audio hours/month across multiple customer tenants.

Jun 2017–Apr 2022 **Software Engineer** → **Lead Software Engineer**, *Samsung Research*, Bangalore, India

- Developed hybrid and end-to-end ASR for accented and far-field speech for Bixby.
- Built VAE-based speaker normalization for far-field ASR; deployed and published at Interspeech.

Technical Skills

Core Python, C++, PyTorch, Hugging Face (Transformers, TRL, PEFT, Datasets), vLLM, fairseq, NeMo, k2/icefall, SpeechBrain, Kaldi

Train at Scale PyTorch DDP, FSDP, DeepSpeed, Slurm, Weights & Biases

Deployment Triton Inference Server, ONNX, Docker, Kubernetes, FastAPI, Flask, gRPC

Professional Activities

Challenges DiSPLACE-M Challenge 2026: Team ranked **1st in 3 of 4 tracks**, including diarization and Hindi Devanagari ASR for medical conversations. [Paper]

Interspeech DiCOVA Challenge 2021: Ranked 5th out of 29 teams in a COVID-19 detection task using cough sounds (AUC 83.93). [Paper]

Reviewer NeurIPS, ICLR, ICASSP, Interspeech, EUSIPCO, and ACL Rolling Review (ARR).

Presentations NeurIPS 2025, Interspeech 2025, and EMNLP 2024.

Awards **Best Paper Award**, SALMA Workshop at ICASSP 2025.

Idiap PhD Paper Award, for "Latent Space Factorization in LoRA" (NeurIPS 2025).